

Shubham Karampure

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Skills

Languages: C, C++, SQL, HTML, CSS, JavaScript

Frameworks/Libraries: React, Next.js, Node.js (w/ Express.js), Flask, PyTorch, Socket.IO, LangChain, CrewAI, TailwindCSS

Databases: PostgreSQL, MongoDB, Redis, Qdrant

Tools: Git, GitHub, RabbitMQ

AI/ML: Generative AI (RAG, LLM Orchestration), Hugging Face (Transformers, Diffusers)

Education

B.Tech in Computer Science and Engineering (Data Science)

2022 – 2026

Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology

CGPA: 8.01/10 Andheri, India

Projects

Eduverse – AI-Assisted Learning Management Platform

Sep 2024

- Engineered AI-powered LMS with ERP features using LangChain, RAG, and Qdrant for intelligent content processing & semantic search.
- Built decoupled GenAI microservice (Python/Flask) for automated personalized roadmaps, quizzes, & Marp/Mermaid presentations via Llama 3.1 (Groq) & Gemini APIs.
- Implemented AI-enhanced grading with instant feedback, customizable rubrics, plagiarism & AI content detection.

Real-time Distributed Scribble Game

May 2025

- Developed distributed multiplayer drawing/guessing game using Node.js, Socket.IO, and MongoDB.
- Engineered real-time synchronized drawing/chat; scaled Socket.IO horizontally with a Redis Pub/Sub adapter.
- Implemented Bully Algorithm for leader election (fault tolerance) & Vector Clocks for causal event-order consistency.
- Utilized RabbitMQ for asynchronous game analytics processing & a gRPC microservice for decoupled word generation.

Devanagari Glyph Generation via Denoising Diffusion Models

April 2025

- Implemented DDPM from scratch using PyTorch for generating Devanagari script characters with custom U-Net architecture.
- Achieved high-quality glyph generation through optimized noise scheduling and diffusion process fine-tuning.

Certifications and Courses

The Complete Web Development Bootcamp – Angela Yu (Udemy)

2025

- Learned full-stack web development: HTML, CSS, JavaScript, Node.js, MongoDB, and React.

Pursuing Minors in Generative AI by Vizvara Labs

Ongoing

- Studying RNNs, LSTMs, attention mechanisms, GANs, and transformers through lectures and coding assignments.

Achievements

- Solved 600+ DSA problems** across multiple programming platforms.
- Hackathon Wins:** Winner of 5 hackathons including N.I.T.-Hacks 7.0 (300+ teams), IEEE HackExplore (200+ teams), Airavat Hackathon (GEN-AI Domain), and finalist in Smart India Hackathon (Top 5 of 500+ submissions).
- Runner-up:** CodeHousie & CodeItOut (Intra-college Competitive Coding).

Leadership Experience

Head of Finance

Jul 2022 – Jun 2024

E-CELL S.P.I.T.

- Managed 6+ lakhs of funding with precise budgeting, GST compliance, and income-expense tracking.
- Led 30+ member teams to execute events like E-Summit (1500+ attendees), cutting costs by 20% through optimized planning.